



Midterm Exam

International Campus Pharma D Program

Biochemistry

September 2023 semester

Name and Surname.

Student ID Number:

Course Coordinator: Dr. Panahi

Date of Exam: Jan 06, 2024

Duration of Exam: 65Minutes

MCQ Question	64
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Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.

1. The chemical reactions in the small intestine take place in a basic (alkaline) environment. Which number indicates this basic pH?

- a. 2.5
- b. 4.6
- c. 6.9
- ☒ d. 8.5

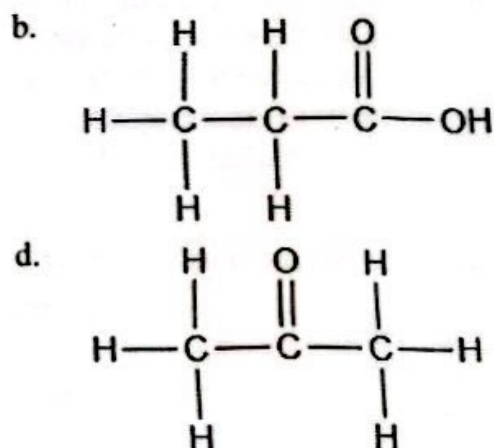
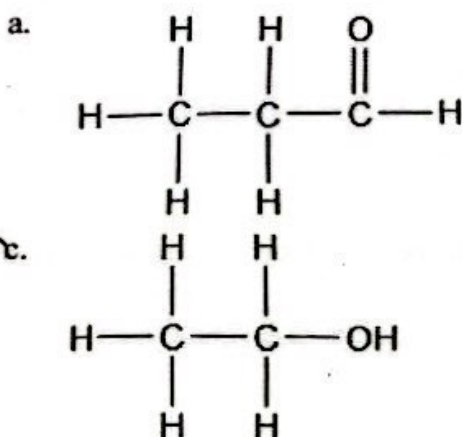
2. Which substance is produced in every dehydration synthesis reaction?

- a. Fat
- b. Water
- ☒ c. Protein
- d. Carbohydrate

3. What happens when ice melts?

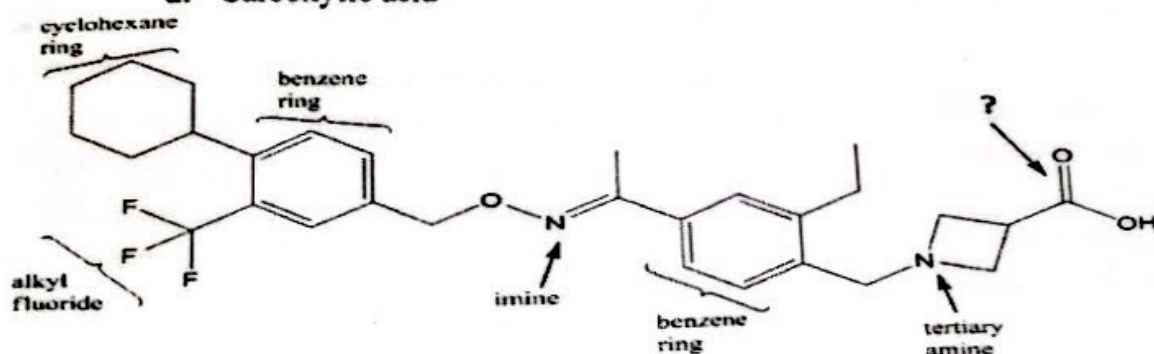
- a. Water molecules change to hydrogen and oxygen atoms.
- b. Water molecules change to water atoms.
- ☒ c. Irregularly arranged molecules change to regularly arranged molecules.
- d. Regularly arranged molecules change to irregularly arranged molecules.

4. Identify the compound which has *alcohol* as the functional group.



5. What is the name of the group shown as question mark in the structure below?

- a. Hydroxyl
- ☒ b. Aldehyde
- c. Ether
- d. Carboxylic acid



6. Aspirin is an acid with a $pK_a = 3.4$. At the stomach $pH = 1.4$, what would be the ratio of A^- to HA .
- a. -2
 - b. +2
 - ☒ c. 100
 - d. 1/100
7. A buffer system can be made of a weak acid and its salt or a weak base and its salt. Which of the following can make a buffer?
- a. H_3PO_4 and Na_3PO_4
 - b. NH_4NO_3 and NH_3
 - c. $NaNO_3$ and $Ca(NO_3)_2$
 - ☒ d. HCl and $NaCl$
8. The primary buffer systems in the urine?
- a. Bicarbonate and hemoglobin
 - ☒ b. Albumin and phosphate
 - c. bicarbonate and ammonia
 - d. Phosphate and ammonia
9. In sickle cell anemia, the basis of the malfunction of the hemoglobin molecule is:
- a) Insufficient iron in the diet
 - b) Substitution of a single amino acid
 - ☒ c) Incorrect secondary structure
 - d) Faulty binding of the heme groups
10. According to the given values, what would be the pI for glutamic acid?
- | | | |
|-----------------|--------------------|--------------------|
| $pK_{aR} = 4.2$ | $pK_{aCOOH} = 2.3$ | $pK_{aNH_2} = 9.3$ |
|-----------------|--------------------|--------------------|
- a) 6.75
 - b) 3.25
 - c) 5.8
 - ☒ d) 4.25
11. Amino acids with the aromatic 'R' group are:
- a) Glycine, alanine, leucine
 - b) Serine, threonine, cysteine
 - c) Lysine, arginine, histidine
 - d) Phenylalanine, tyrosine and tryptophan
12. β -pleated sheets are the examples of?
- ☒ a) Secondary structure
 - b) Tertiary structure
 - c) Primary structure
 - d) Quaternary structure

13. Which of the following factor increase the hemoglobin affinity for oxygen?

- a) Increase of CO₂ pressure
- b) Increase of CO
- c) Decrease of pH
- d) Increase of temperature

14. All of the following statement about Myoglobin is correct, except:

- a) Consist of a single polypeptide chain.
- b) Its secondary structure is rich in β -sheet.
- c) Is a compact and spherical molecule.
- d) Present in heart and skeletal muscle

15. What are the most common amino acids in collagen?

- a) Tryptophan and Prolin
- b) Glycine and Lysine
- c) Threonine and Proline
- d) Proline and Glycine

16. Which of the following proteins is defective in Marfan syndrome?

- a) Collagen
- b) Elastase
- c) Fibrillin
- d) Hemoglobin

17. Which of the following reaction is catalyzed by isomerase?

- a) Formation of bonds
- b) Breaking of bonds
- c) Intramolecular rearrangement of bonds
- d) Transfer of group from one molecule to another

18. According to the Michaelis-Menten Kinetics, when substrate concentration is very high:

- a) $V_0 = 2 V_{max}$
- b) $V_0 = V_{max}/2$
- c) $V_0 = V_{max}$
- d) $V_0 = V_{max}/4$

19. The general mechanism is that an enzyme acts by?

- a) Increasing the free energies of the reactants
- b) Reducing the activation energy
- c) Decreasing the free energies of the products
- d) Increasing activation energy

20. In Lineweaver Burk plots, the y-intercept is equal to:

- a) V_{max}/K_m
- ☒ b) $1/V_{max}$
- c) $-1/K_m$
- d) V_{max}

21. Which statement is incorrect about allosteric enzymes?

- a) Display a sigmoidal dependence on substrate concentration
- b) Frequently catalyze the committed step early in a pathway
- c) Are almost always composed of multiple subunits
- ☒ d) An allosteric substrate functions as a negative effector

22. Which of the following is not an example for competitive inhibitor?

- ☒ a) Methotrexate for dihydrofolate reductase
- b) Atorvastatin for HMG-CoA reductase
- c) Amoxicillin for transpeptidase
- d) Malonate for succinate dehydrogenase

23. Which of the following statement about isoenzymes is not true:

- a) Display different regulatory properties
- ☒ b) Have distinct physical properties
- c) Catalyze the different reaction
- d) May be expressed in a tissue-specific

24. In non-competitive inhibition, what happens to V_{max} and K_m ?

- a) K_m increased but V_{max} decreased
- b) V_{max} increased but K_m is constant
- ☒ c) K_m is constant but V_{max} decreased
- d) V_{max} decreased but K_m increased

25. Riboflavin deficiency causes all except:

- a) Cheilosis
- b) Glossitis
- ☒ c) Nystagmus
- d) Dermatitis

26. Which of the following vitamins are water soluble?

- ☒ a) Vitamin B and Vitamin C
- b) Vitamin A and Vitamin B
- c) Vitamin K and Vitamin D
- d) Vitamin C and Vitamin E

27. Which of the following diseases is caused by the deficiency of Niacin?
- a) Scurvy
 - b) Rickets
 - ☒ c) Pellagra
 - d) Pernicious anemia
28. Which of the following is effective in premenstrual syndrome (PMS) and diabetes?
- a) Vitamin B12
 - b) Vitamin B9
 - c) Vitamin B6
 - d) Vitamin B3
29. Which of these is a vitamin A precursor?
- a) Thiamine
 - b) Pyridoxine
 - ☒ c) Cobalamin
 - d) Beta-carotene
30. Which of the following nutrient deficiency causes megaloblastic anemia?
- a) Pantothenic acid
 - ☒ b) Niacin
 - c) Folic acid
 - d) Retinol
31. Which of the following organs are responsible for the activation of vitamin D in the human body?
- a) Intestine and Liver
 - b) Kidney and Pancreas
 - ☒ c) Liver and Kidney
 - d) Stomach and Brain
32. The most biologically active form of vitamin K is:
- a) Menadione
 - b) Menaquinone
 - ☒ c) Phylloquinone
 - d) Phytonadion
33. Which of the following monosaccharides is the precursor for synthesis of glycogen as a storage carbohydrate in animals?
- a) Fructose
 - b) Glucose
 - ☒ c) Ribose
 - d) Mannose

46. **Prostaglandins are synthesized in the body from**

- a) Myristic acid
- ✓ b) Arachidonic acid
- c) Stearic acid
- d) Lignoceric acid

47. **Dietary fats after absorption appear in the circulation as**

- a) Chylomicron
- b) VLDL
- ✓ c) LDL
- d) HDL

48. **A fatty acid which is not synthesized in the body and has to be supplied in the diet is**

- a) Palmitic acid
- b) Lauric acid
- ✓ c) Linolenic acid
- d) Palmitoleic acid

49. **An increased melting temperature of duplex DNA results from a high content of**

- a) Adenine + Guanine
- ✓ b) Thymine + Cytosine
- c) Cytosine + Guanine
- d) Cytosine + Adenine

50. **The enzyme DNA ligase**

- a) Introduces superhelical twists
- b) Connects the end of two DNA chains
- ✓ c) Unwinds the double helix
- d) Synthesises RNA primers

51. **RNA primer is formed by the enzyme:**

- a) Ribonuclease
- b) DNA polymerase I
- ✓ c) DNA polymerase III
- d) Primase

52. **Ciprofloxacin inhibits**

- a) DNA topoisomerase I
- b) DNA polymerase I
- ✓ c) DNA polymerase III
- d) DNA gyrase

39. One dermatologists want to design a substance that is used to prevent and treat blood in blood vessels and the heart. Which of the following carbohydrates will be suggested?
- ☒ a) Heparin
 - b) Keratan sulphate
 - c) Chondrotin sulphate
 - d) Heparan sulphate
40. Which biomolecule can be an appropriate candidate for control of calcification in bone?
- a) Hyaluronic acid
 - ☒ b) Keratan sulphate
 - c) Chondrotin sulphate
 - d) Dermatan sulphate
41. Which is a characteristic of sphingolipids?
- ☒ a) They all contain a fatty acid joined to glycerol.
 - b) They all contain a long-chain alcohol joined to isoprene
 - c) They all contain ceramide joined to a polar group
 - d) They all contain a carbohydrate joined to a phosphate group.
42. The nitrogenous base in lecithin is
- a) Ethanolamine
 - ☒ b) Serine
 - c) Betaine
 - d) Choline
43. Which of the following is a polyunsaturated fatty acid?
- a) Oleic acid
 - b) Stearic acid
 - ☒ c) Linoleic acid
 - d) Palmitoleic acid
44. Very low-density lipoproteins are also known as
- a) β -lipoproteins
 - ☒ b) Pre β -lipoproteins
 - c) α -lipoprotein
 - d) Broad β -lipoproteins
45. The lipoprotein richest in cholesterol is
- a) Chylomicron
 - b) VLDL
 - c) LDL
 - ☒ d) HDL

34. α -D-glucose and β -D-glucose are:

- a) Stereoisomer
- b) Epimers
- ☒ c) Anomers
- d) Keto-Aldo pairs

35. Which of the following are enantiomers:

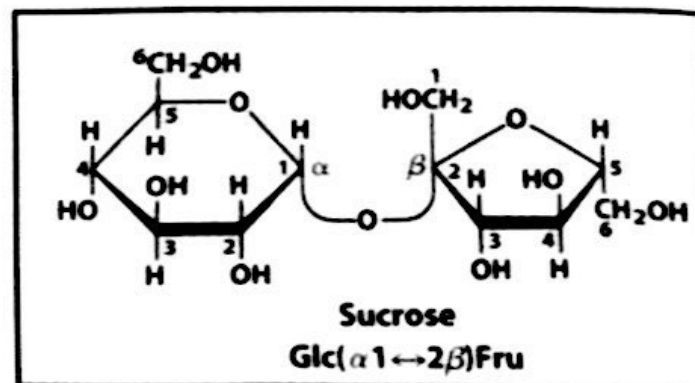
- a) D-Mannose and L-Glucose
- ☒ b) d- Galactose and l-Glucose
- c) D-Galactose and L-Galactose
- d) d-Mannose and l-Mannose

36. Which of the following monosaccharides can act as a structural component for biosynthesis of ATP?

- a) Mannose
- ☒ b) Ribose
- c) Arabinose
- d) Ribulose

37. The following structure is related to sucrose. Which statement is true about it?

- a) Reducing disaccharides
- b) Nutritionally important disaccharide
- ☒ c) It is composed of two aldohexose
- d) It can be found mostly in milk



38. Which statement is true about amylose and amylopectin?

- a) Both of them contain alpha linkages
- ☒ b) Amylose contain alpha linkage but amylopectin contains beta linkage
- c) Amylopectin has a linear structure but amylose contains branched chain
- d) Both of them contain branched chain

53. In RNA molecule 'Caps'

- a) Allow tRNA to be processed
- ☒ b) Occur at the 3' end of mRNA
- c) Allow correct translation of prokaryotic mRNA
- d) Are unique to eukaryotic mRNA

54. Reverse transcriptase catalyses

- a) Synthesis of RNA
- b) Breakdown of RNA
- ☒ c) Synthesis of DNA
- d) Breakdown of DNA

55. The initiation site for transcription is recognized by

- ☒ a) Sigma factor
- b) Rho factor
- c) β -Subunit of DNA-dependent RNA polymerase
- d) α -Subunit of DNA-dependent RNA polymerase

56. In prokaryotes, chloramphenicol

- a) Preventing binding of aminoacyl tRNA
- b) Causes misreading of the mRNA
- c) Depolymerises DNA
- ☒ d) Inhibits peptidyl transferase activity

57. Which of these operons is anabolic?

- a) Lac
- b) Ara
- ☒ c) Trp
- d) Phe

58. Xeroderma pigmentosum results from a defect in

- a) uvr ABC excinuclease
- b) DNA polymerase I
- ☒ c) DNA ligase
- d) DNA polymerase III

59. If the codon UAC on mRNA changes into UAG as a result of a base substitution in DNA, it will result in

- ☒ a) Silent mutation
- b) Acceptable mis-sense mutation
- c) Nonsense mutation
- d) Frameshift mutation

60. Transcription of z, y and a genes of lac operon is prevented by

- ☒ a) Repressor
- b) Lactose
- c) Allo-lactose
- d) cAMP

61. The 'rho' (ρ) factor is involved

- a) To increase the rate of RNA synthesis
- b) In binding catabolite repressor to the promoter region
- ☒ c) In proper termination of transcription
- d) To allow proper initiation of transcriptide

62. The anticodon region is an important part of the structure of

- ☒ a) rRNA
- b) tRNA
- c) mRNA
- d) hrRNA

63. The antibiotic which cleaves DNA is

- a) Actinomycin d
- ☒ b) Puromycin
- c) Streptomycin
- d) Mitomycin C

64. A splice site contains all the following except

- a) Ribosome
- ☒ b) Some proteins
- c) hnRNA
- d) snRNAs

pharmacy term 2
Biochemistry answer key
(Midterm)

Attention! 1- If this answer-sheet is not yours, Please inform the responsible person
2- The answers should be marked with a soft and bold black pencil in the appropriate box.

How to Mark: Wrong ☐ ☐ ☐ ☐ Correct ☒

Please don't write anything in this box.

1 ☐ a ☐ b ☐ c ☒ d	41 ☐ a ☐ b ☒ c ☐ d	81 ☐ a ☐ b ☐ c ☐ d	121 ☐ a ☐ b ☐ c ☐ d
2 ☐ a ☒ b ☐ c ☐ d	42 ☐ a ☐ b ☐ c ☒ d	82 ☐ a ☐ b ☐ c ☐ d	122 ☐ a ☐ b ☐ c ☐ d
3 ☐ a ☐ b ☐ c ☒ d	43 ☐ a ☐ b ☒ c ☐ d	83 ☐ a ☐ b ☐ c ☐ d	123 ☐ a ☐ b ☐ c ☐ d
4 ☐ a ☐ b ☒ c ☐ d	44 ☐ a ☒ b ☐ c ☐ d	84 ☐ a ☐ b ☐ c ☐ d	124 ☐ a ☐ b ☐ c ☐ d
5 ☐ a ☐ b ☐ c ☒ d	45 ☐ a ☐ b ☒ c ☐ d	85 ☐ a ☐ b ☐ c ☐ d	125 ☐ a ☐ b ☐ c ☐ d
6 ☐ a ☐ b ☐ c ☒ d	46 ☐ a ☒ b ☐ c ☐ d	86 ☐ a ☐ b ☐ c ☐ d	126 ☐ a ☐ b ☐ c ☐ d
7 ☐ a ☒ b ☐ c ☐ d	47 ☒ a ☐ b ☐ c ☐ d	87 ☐ a ☐ b ☐ c ☐ d	127 ☐ a ☐ b ☐ c ☐ d
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10 ☐ a ☒ b ☐ c ☐ d	50 ☐ a ☒ b ☐ c ☐ d	90 ☐ a ☐ b ☐ c ☐ d	130 ☐ a ☐ b ☐ c ☐ d
11 ☐ a ☐ b ☐ c ☒ d	51 ☐ a ☐ b ☐ c ☒ d	91 ☐ a ☐ b ☐ c ☐ d	131 ☐ a ☐ b ☐ c ☐ d
12 ☒ a ☐ b ☐ c ☐ d	52 ☐ a ☐ b ☐ c ☒ d	92 ☐ a ☐ b ☐ c ☐ d	132 ☐ a ☐ b ☐ c ☐ d
13 ☐ a ☒ b ☐ c ☐ d	53 ☐ a ☐ b ☐ c ☒ d	93 ☐ a ☐ b ☐ c ☐ d	133 ☐ a ☐ b ☐ c ☐ d
14 ☐ a ☒ b ☐ c ☐ d	54 ☐ a ☐ b ☒ c ☐ d	94 ☐ a ☐ b ☐ c ☐ d	134 ☐ a ☐ b ☐ c ☐ d
15 ☐ a ☐ b ☐ c ☒ d	55 ☒ a ☐ b ☐ c ☐ d	95 ☐ a ☐ b ☐ c ☐ d	135 ☐ a ☐ b ☐ c ☐ d
16 ☐ a ☐ b ☒ c ☐ d	56 ☐ a ☐ b ☐ c ☒ d	96 ☐ a ☐ b ☐ c ☐ d	136 ☐ a ☐ b ☐ c ☐ d
17 ☐ a ☐ b ☐ c ☒ d	57 ☐ a ☐ b ☒ c ☐ d	97 ☐ a ☐ b ☐ c ☐ d	137 ☐ a ☐ b ☐ c ☐ d
18 ☐ a ☐ b ☒ c ☐ d	58 ☒ a ☐ b ☐ c ☐ d	98 ☐ a ☐ b ☐ c ☐ d	138 ☐ a ☐ b ☐ c ☐ d
19 ☐ a ☒ b ☐ c ☐ d	59 ☐ a ☐ b ☒ c ☐ d	99 ☐ a ☐ b ☐ c ☐ d	139 ☐ a ☐ b ☐ c ☐ d
20 ☐ a ☒ b ☐ c ☐ d	60 ☒ a ☐ b ☐ c ☐ d	100 ☐ a ☐ b ☐ c ☐ d	140 ☐ a ☐ b ☐ c ☐ d
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22 ☐ a ☐ b ☒ c ☐ d	62 ☐ a ☒ b ☐ c ☐ d	102 ☐ a ☐ b ☐ c ☐ d	142 ☐ a ☐ b ☐ c ☐ d
23 ☐ a ☐ b ☐ c ☒ d	63 ☐ a ☐ b ☐ c ☒ d	103 ☐ a ☐ b ☐ c ☐ d	143 ☐ a ☐ b ☐ c ☐ d
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25 ☐ a ☐ b ☒ c ☐ d	65 ☐ a ☐ b ☐ c ☐ d	105 ☐ a ☐ b ☐ c ☐ d	145 ☐ a ☐ b ☐ c ☐ d
26 ☒ a ☐ b ☐ c ☐ d	66 ☐ a ☐ b ☐ c ☐ d	106 ☐ a ☐ b ☐ c ☐ d	146 ☐ a ☐ b ☐ c ☐ d
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30 ☐ a ☐ b ☒ c ☐ d	70 ☐ a ☐ b ☐ c ☐ d	110 ☐ a ☐ b ☐ c ☐ d	150 ☐ a ☐ b ☐ c ☐ d
31 ☐ a ☐ b ☒ c ☐ d	71 ☐ a ☐ b ☐ c ☐ d	111 ☐ a ☐ b ☐ c ☐ d	151 ☐ a ☐ b ☐ c ☐ d
32 ☐ a ☒ b ☐ c ☐ d	72 ☐ a ☐ b ☐ c ☐ d	112 ☐ a ☐ b ☐ c ☐ d	152 ☐ a ☐ b ☐ c ☐ d
33 ☐ a ☒ b ☐ c ☐ d	73 ☐ a ☐ b ☐ c ☐ d	113 ☐ a ☐ b ☐ c ☐ d	153 ☐ a ☐ b ☐ c ☐ d
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35 ☐ a ☐ b ☒ c ☐ d	75 ☐ a ☐ b ☐ c ☐ d	115 ☐ a ☐ b ☐ c ☐ d	155 ☐ a ☐ b ☐ c ☐ d
36 ☐ a ☒ b ☐ c ☐ d	76 ☐ a ☐ b ☐ c ☐ d	116 ☐ a ☐ b ☐ c ☐ d	156 ☐ a ☐ b ☐ c ☐ d
37 ☐ a ☒ b ☐ c ☐ d	77 ☐ a ☐ b ☐ c ☐ d	117 ☐ a ☐ b ☐ c ☐ d	157 ☐ a ☐ b ☐ c ☐ d
38 ☒ a ☐ b ☐ c ☐ d	78 ☐ a ☐ b ☐ c ☐ d	118 ☐ a ☐ b ☐ c ☐ d	158 ☐ a ☐ b ☐ c ☐ d
39 ☒ a ☐ b ☐ c ☐ d	79 ☐ a ☐ b ☐ c ☐ d	119 ☐ a ☐ b ☐ c ☐ d	159 ☐ a ☐ b ☐ c ☐ d
40 ☐ a ☐ b ☒ c ☐ d	80 ☐ a ☐ b ☐ c ☐ d	120 ☐ a ☐ b ☐ c ☐ d	160 ☐ a ☐ b ☐ c ☐ d